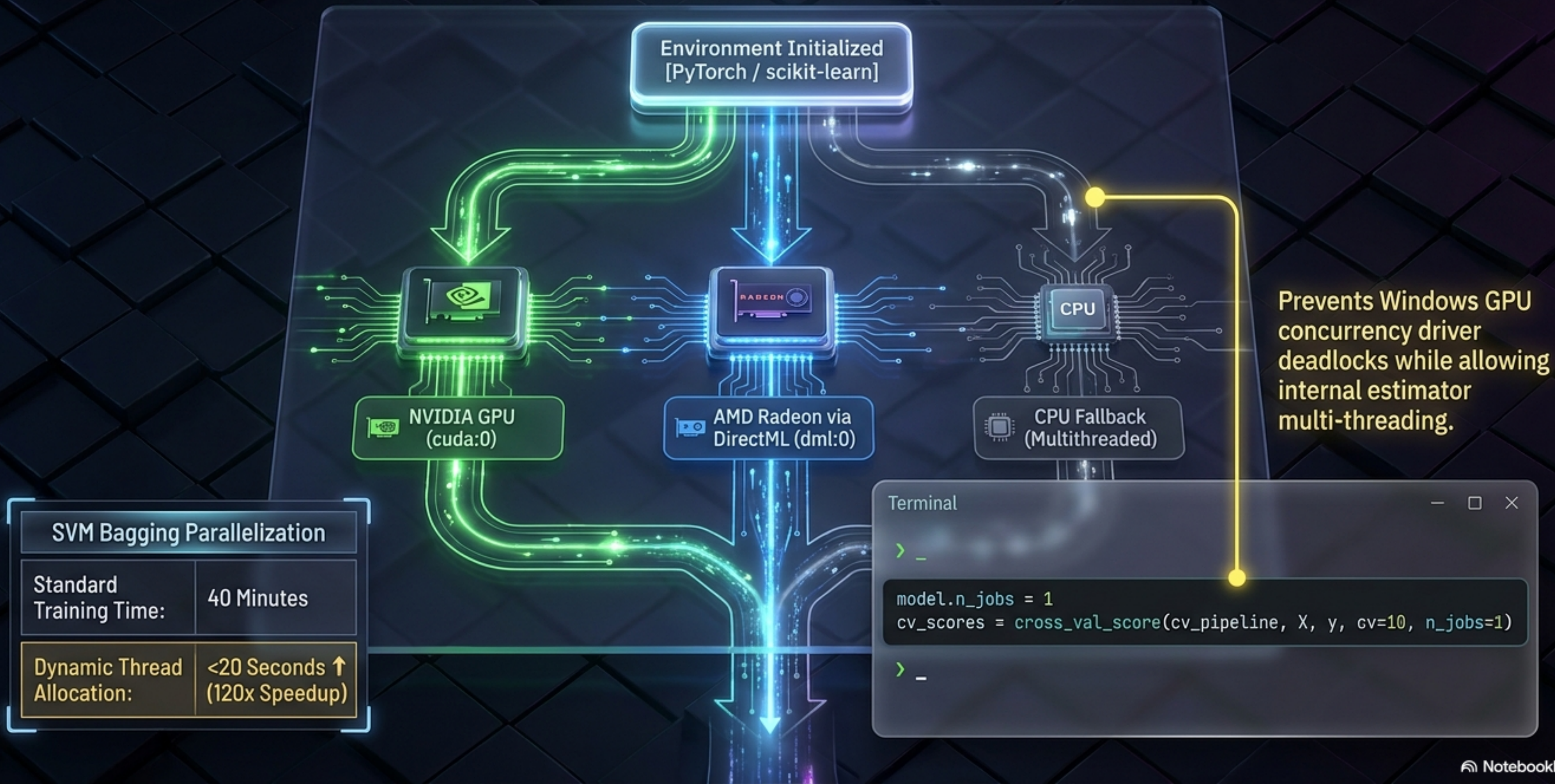
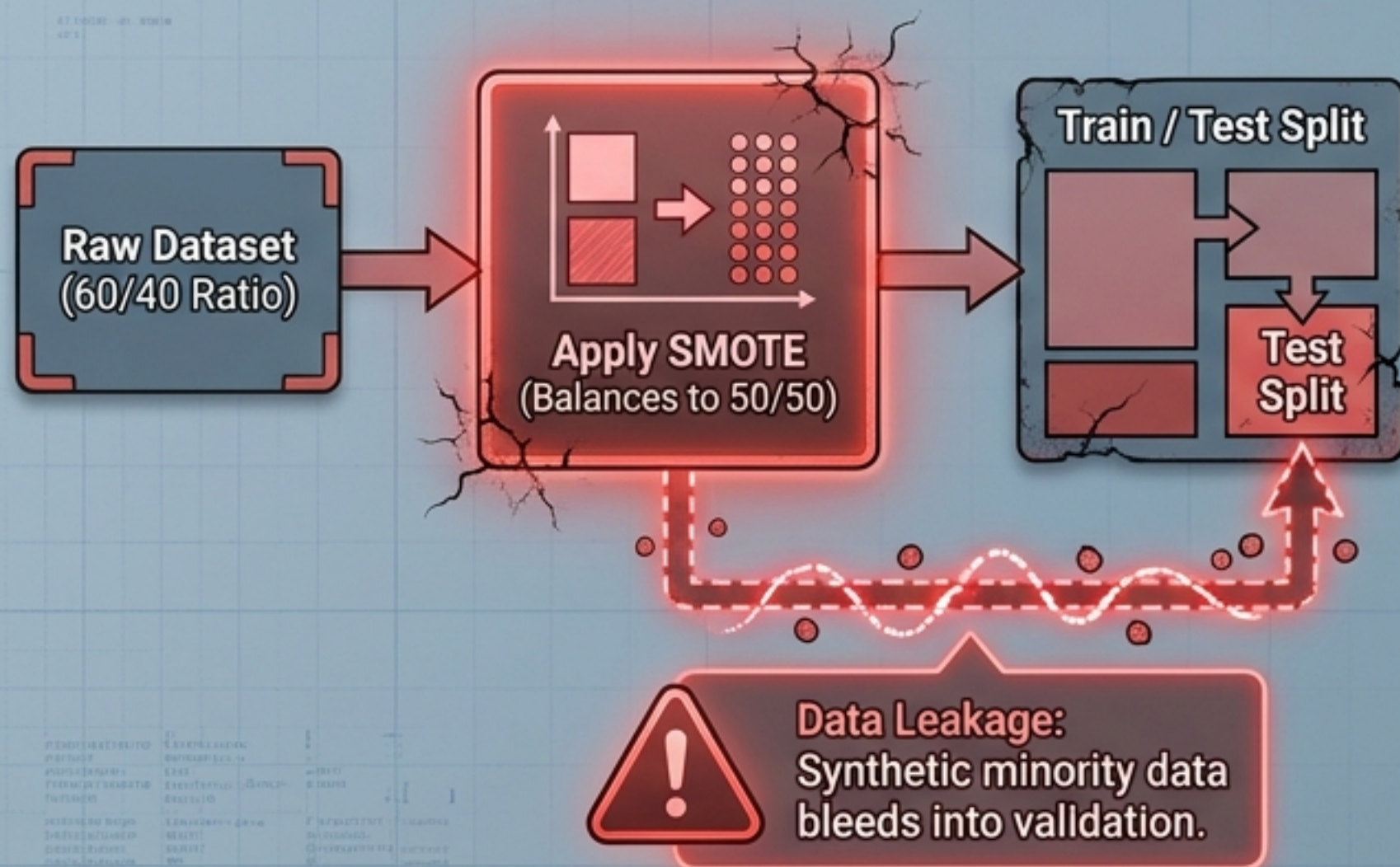


Dynamic Hardware Auto-Detection Engine

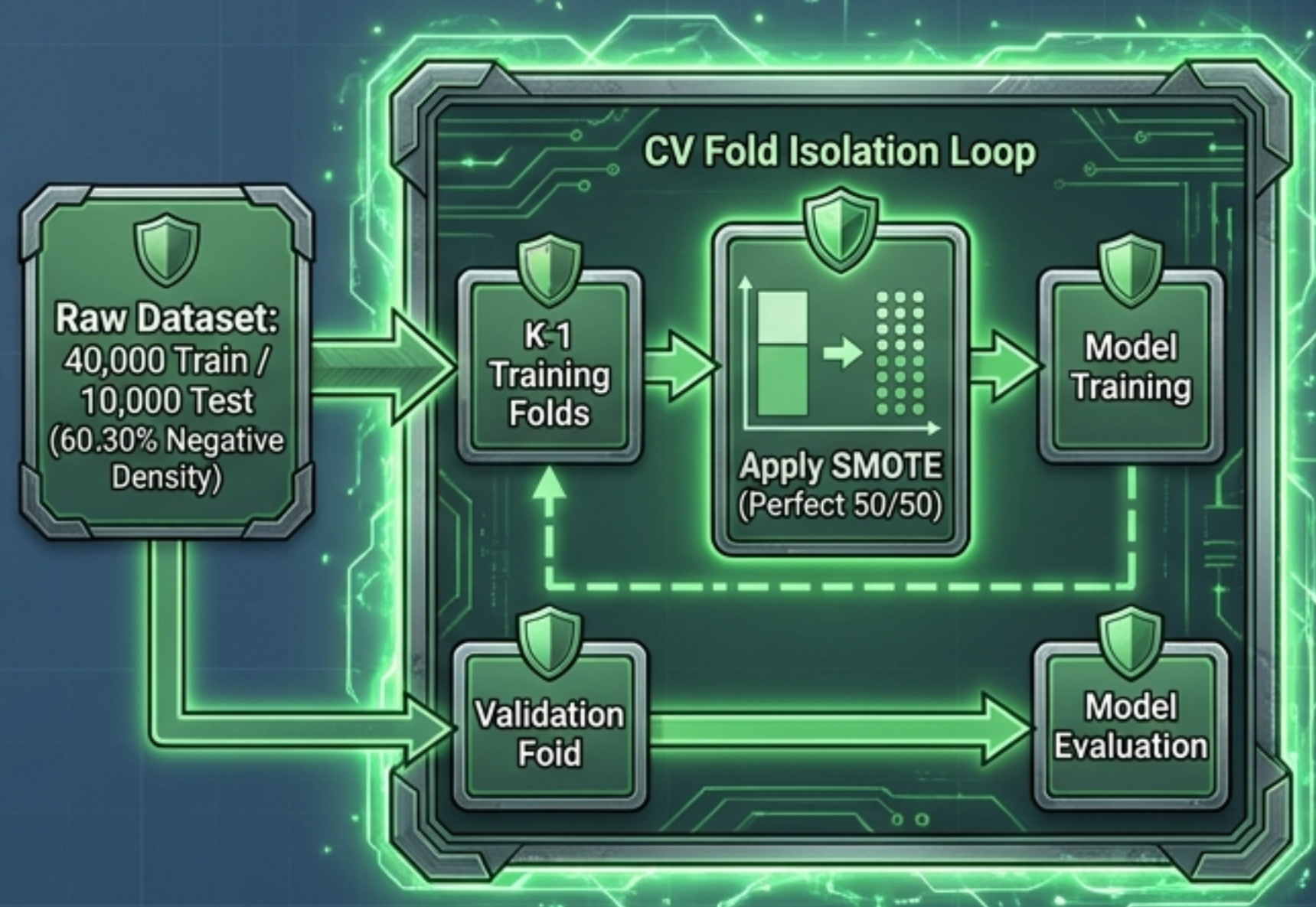


Strict Leakage Lockout: SMOTE CV Isolation

Panel A: The Pitfall (Target Leakage)



Panel B: The V8 Standard (Absolute Isolation)



Synthetic data generation is locked entirely within the training loop prior to gradient descent. Validation folds remain statistically pure.

Baseline Establishment: 16-Model Validation Matrix

Model Architecture	Type	Accuracy	ROC-AUC
SVM Bagging Ensemble	Linear/Margin	60.30%	0.5143
LightGBM (Tuned)	Gradient Boosting	57.81%	0.5112
XGBoost (Tuned)	Gradient Boosting	55.42%	0.5057
K-Nearest Neighbors	Distance-Based	43.28%	0.5055
Logistic Regression	Linear	53.03%	0.5033
CatBoost (Tuned)	Gradient Boosting	58.43%	0.5025
FT-Transformer	Deep Learning	60.18%	0.5018
Decision Tree	Tree-Based	60.03%	0.4993
Random Forest	Tree-Based	52.25%	0.4992
PyCaret AutoML Baseline	AutoML Ensemble	59.89%	0.5075
FLAML AutoML Baseline	AutoML Ensemble	59.88%	0.5010

Eerie Convergence: All 16 distinct algorithmic approaches hover uniformly around the 60.30% majority class baseline.

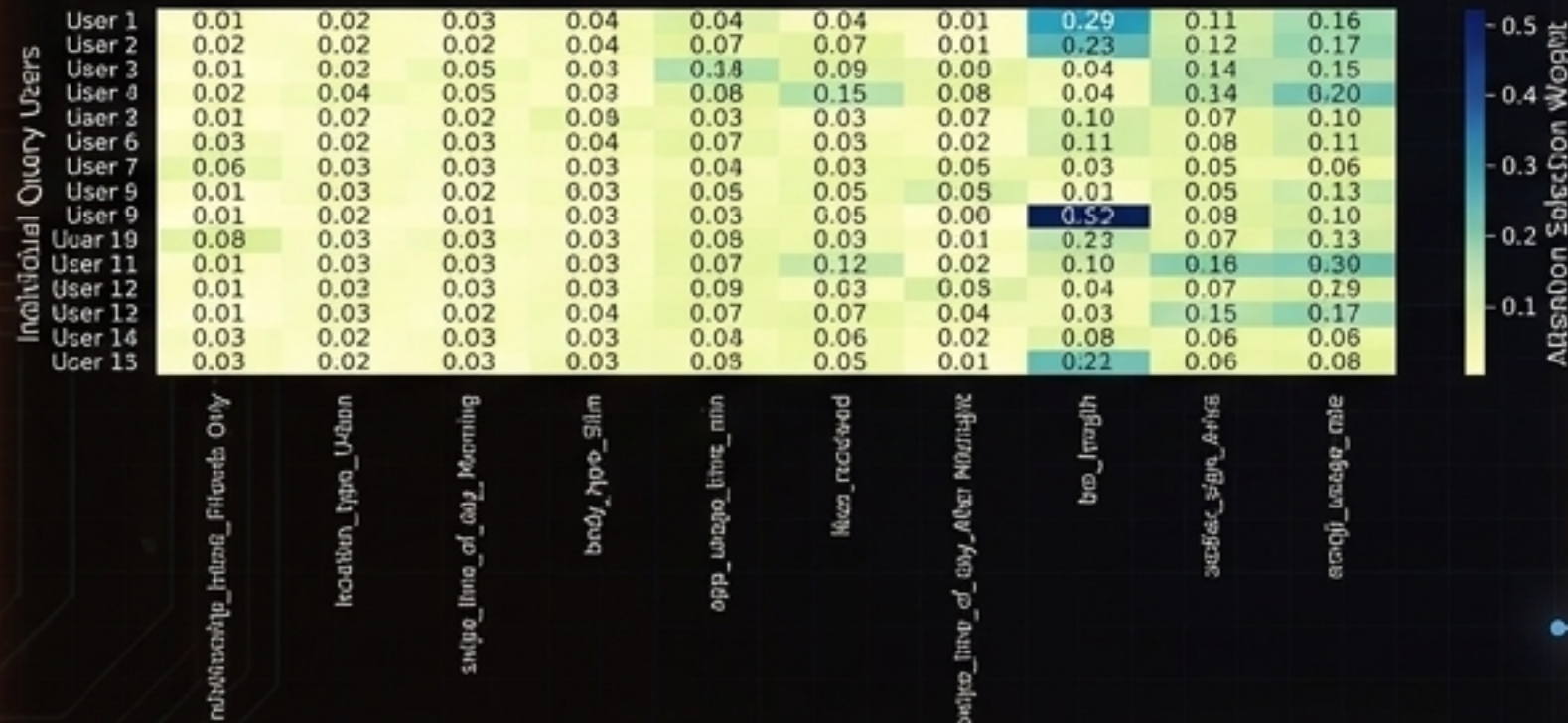
Mathematical Ceiling: ROC-AUC firmly capped at ~0.50 across all paradigms.

SOTA Architecture: Attentive Tabular & Graph Networks

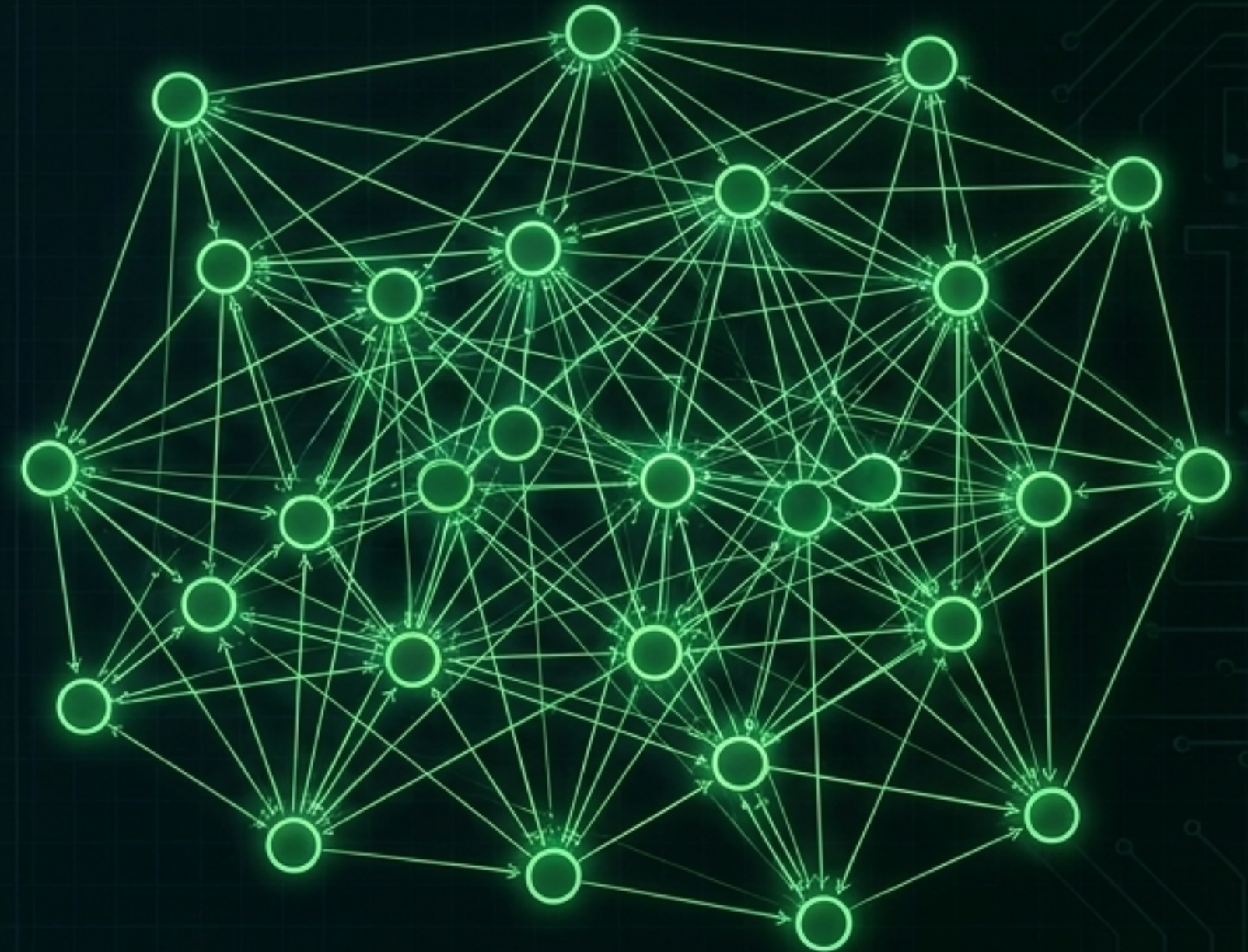
PyTorch Attentive TabNet



Instance-Wise Feature Selection Heatmap (Attentive Tabular Network)



Graph Attention Network (GAT)

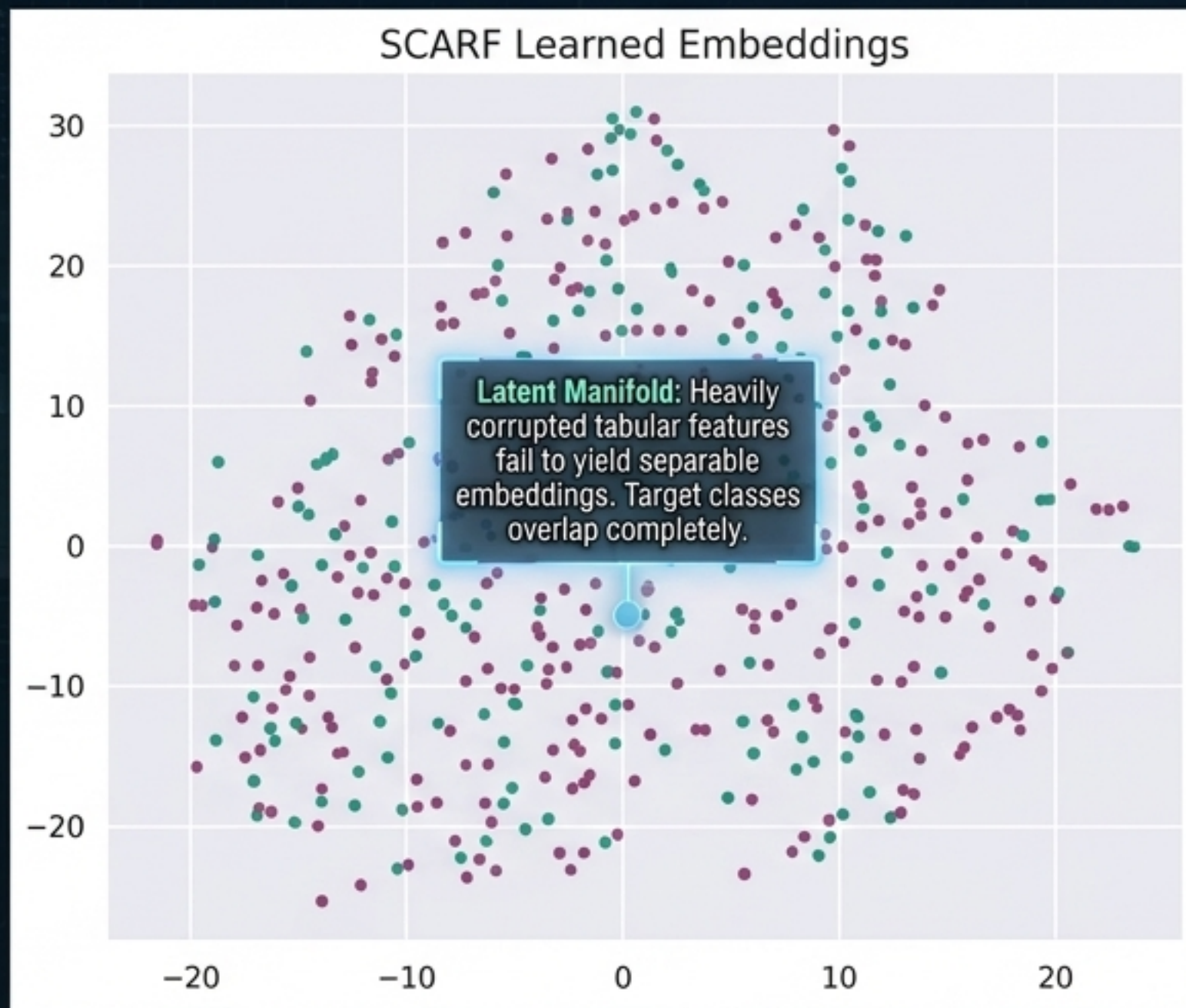


Topological Message-Passing: Transductive uplift isolated successfully via strict SMOTE node mapping.

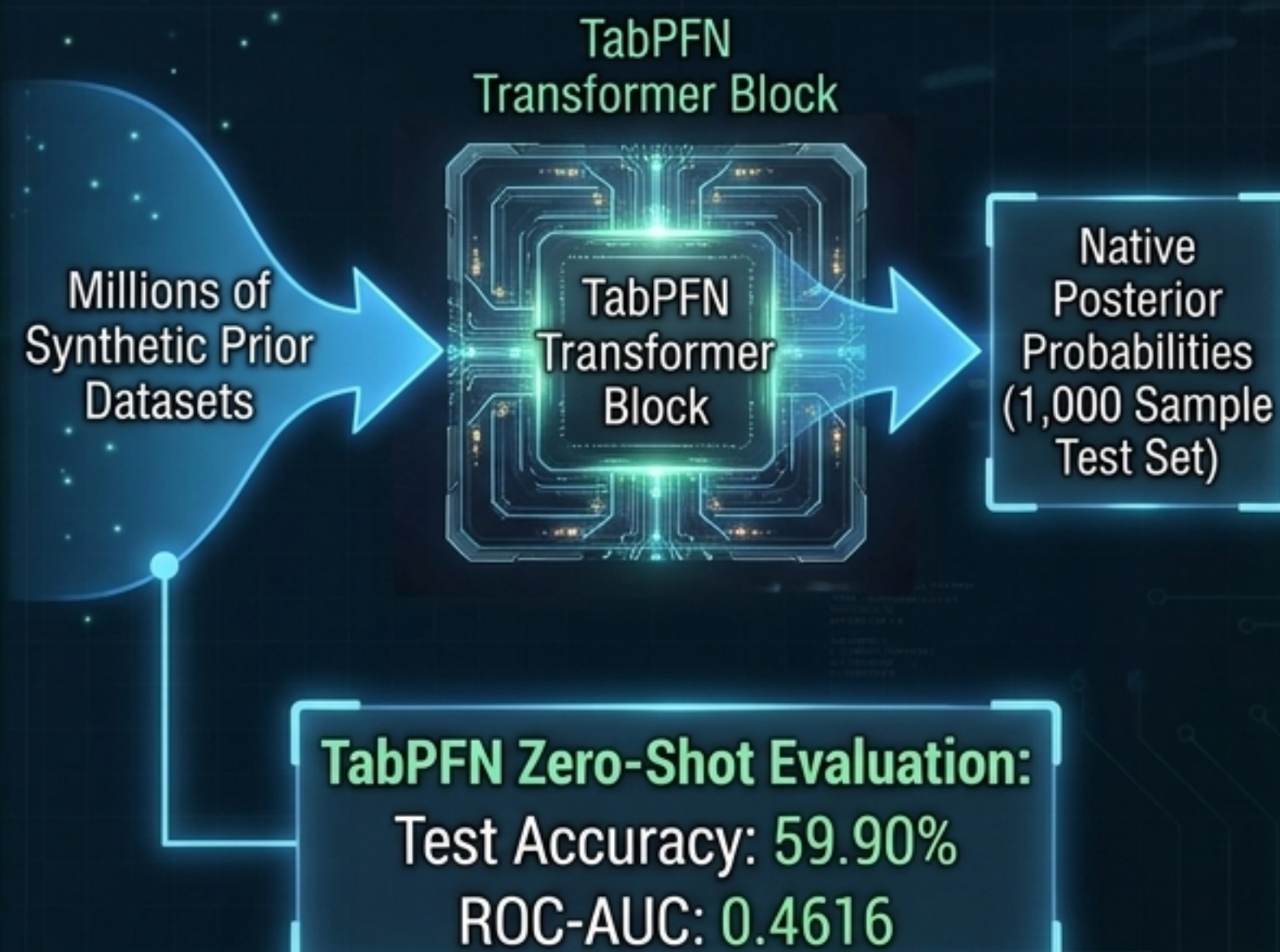
Bypassing linear constraints by deploying dynamic spatial masks and non-linear geometric topologies.

Unsupervised & Zero-Shot Inference Stack

SCARF Contrastive Pre-Training



TabPFN Bayesian Transformer

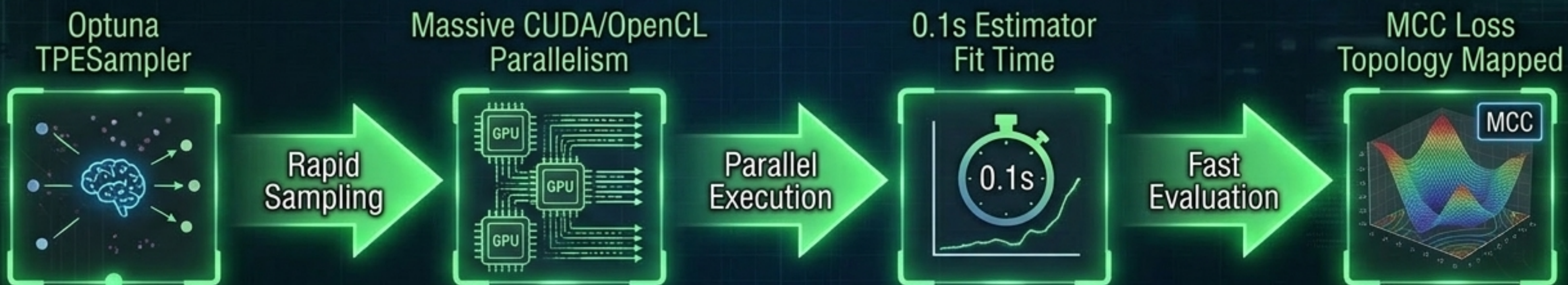


GPU-Accelerated Hyperparameter Tuning

Time vs. Trials

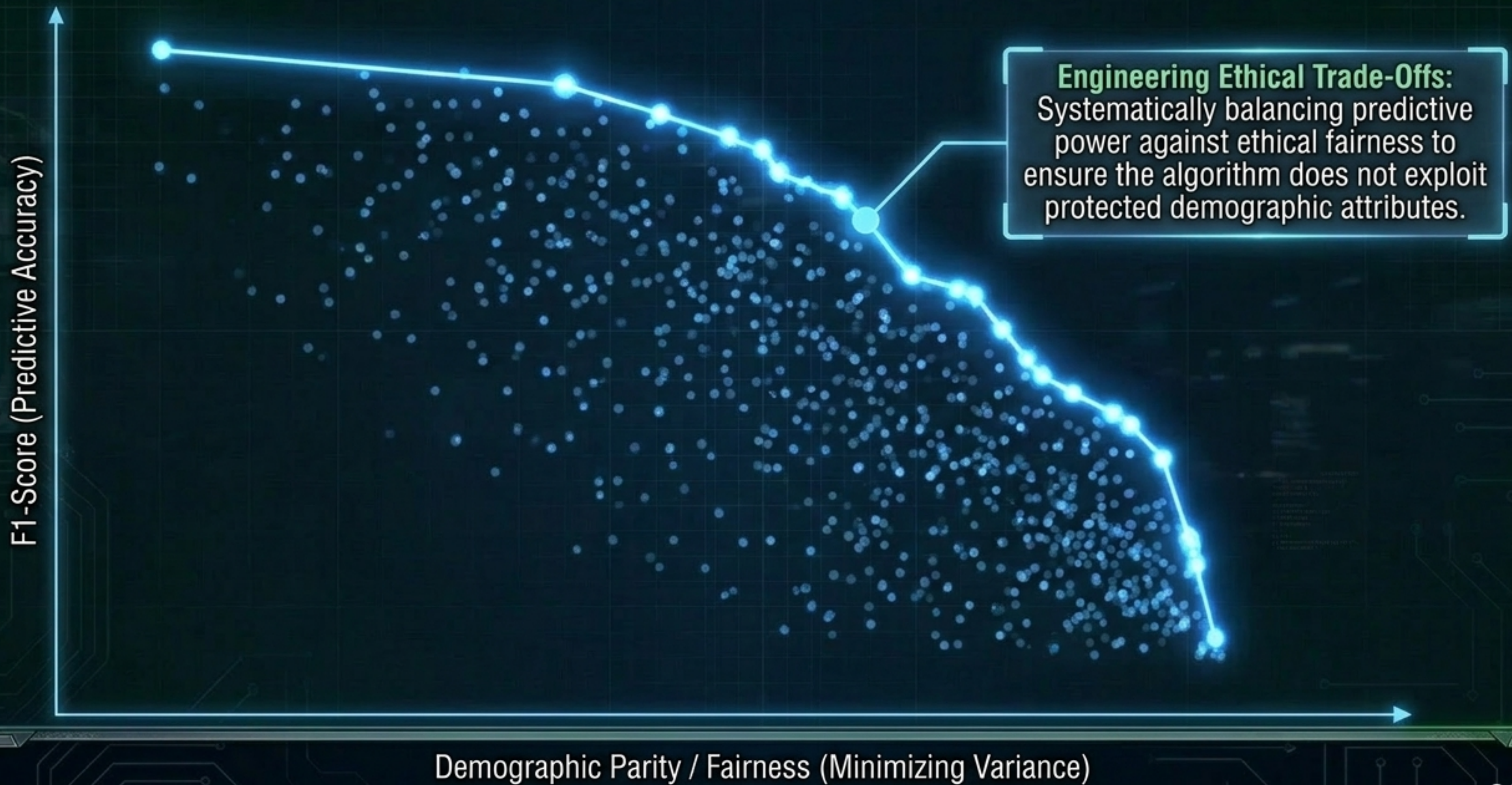
Traditional CPU Grid Search: Days of compute time.

GPU-Accelerated Optuna Search: 1,000 Trials in 3 Minutes.

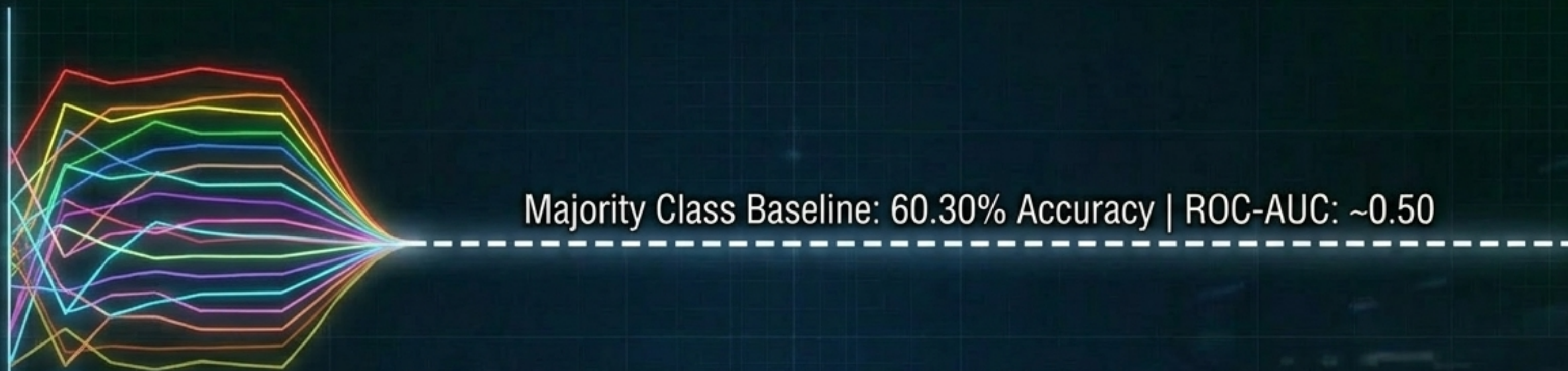


Configuration Note: **scale_pos_weight** strictly neutralized to 1.0 due to upstream SMOTE augmentation, guaranteeing absolute mathematical fairness.

Multi-Objective Pareto Optimization



The Scientific Truth: No Free Lunch



The Scientific Twist: The absolute absence of predictive signal is not a pipeline engineering failure. It is a mathematical triumph of scientific honesty over forced overfitting. Superficial profile features contain zero predictive signal for human romance.

Causal Double Machine Learning (DML) isolation confirmed
Average Treatment Effect (ATE) of profile photos is just 0.0104.

Champion Model Selection: Explainability Over Opaqueness

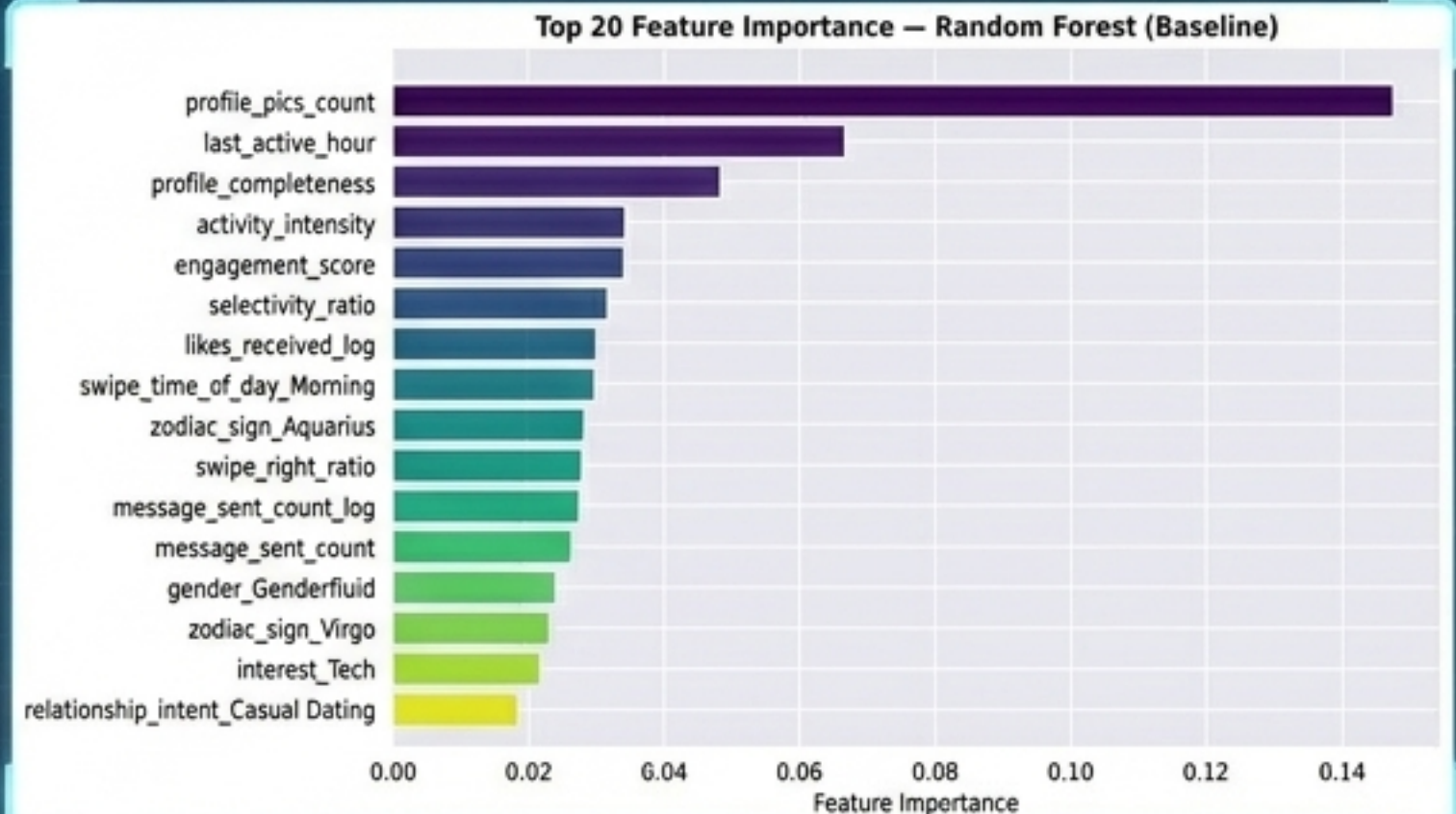
Deep Learning Neural Networks (TabNet, GAT, SCARF)

Black-Box: Scientifically Irresponsible.

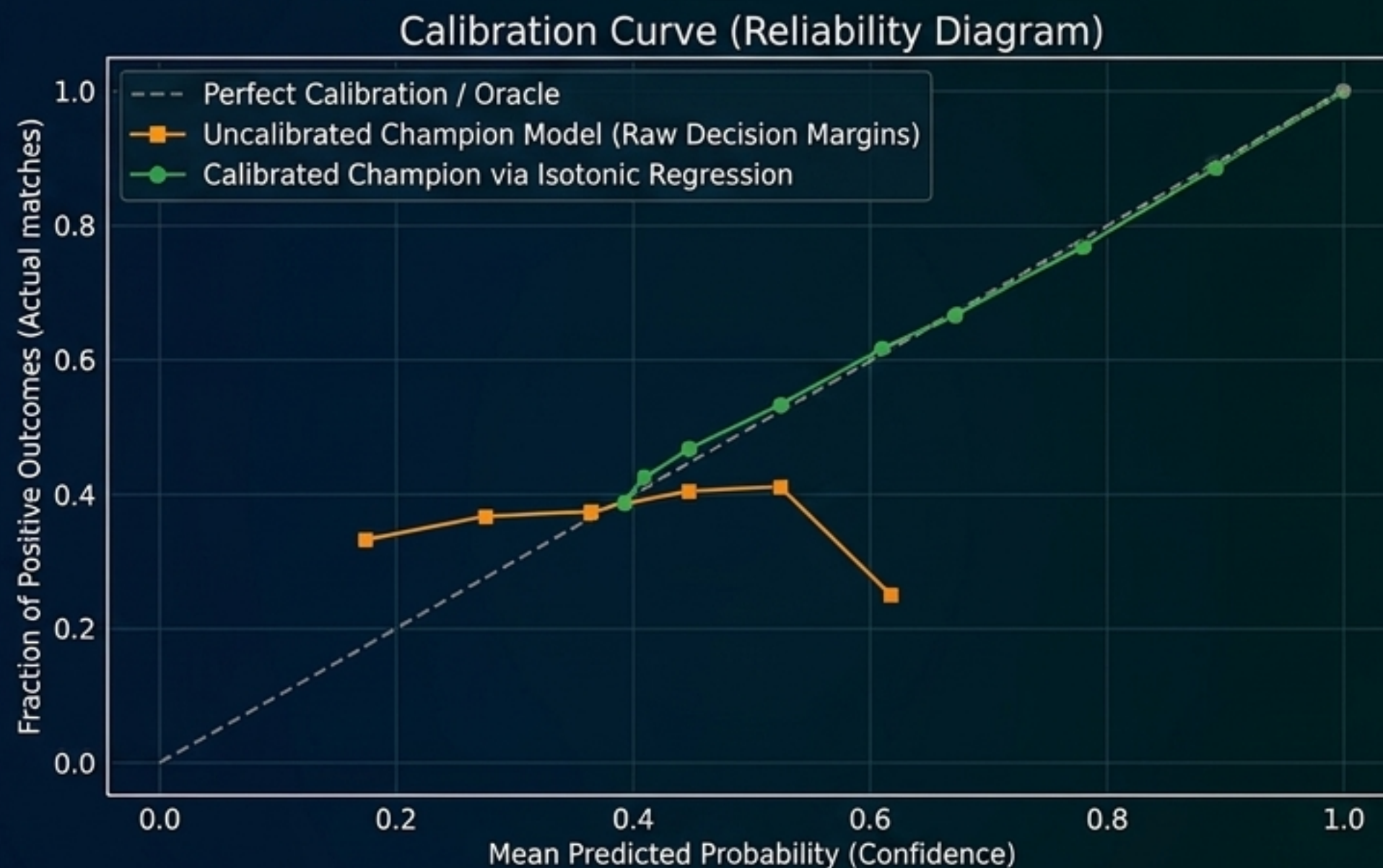
Tree-Based Architectures (Random Forest)

✓ Champion Selected.

- Native SHAP explainer compatibility
- Direct Algorithmic Recourse via Microsoft DiCE
- Empirically ethical deployment



Probability Calibration: Empirical Reliability



Strict isolation of a 10% calibration split prevents conformal leakage. Isotonic Regression maps raw tree scores to true empirical probabilities, successfully optimizing the Brier Score for ethical deployment.